

Modern Physics

Questions with Fully Worked Answers · FormulaVerse PYQ Bank

Q1. For a given circuit containing an ideal diode and a Zener diode ($V_Z = 5V$) between points X and Y, the time-dependent input voltage $v_{in}(t)$ varies as a sine wave between $-20V$ and $+20V$, and the corresponding output $v(t)$ is a signal clamped between $+5V$ and $-5V$. Identify the components used in the circuit between X and Y.

NTA B.Tech · 2 Apr 2026, Shift 1 (Q37)

✓ **Answer: Unverified — see note below**

Solution:

The physics: an ideal diode in series with a 5V Zener diode (oriented to clip both positive and negative excursions) produces exactly this kind of symmetric $\pm 5V$ clamped output from a $\pm 20V$ input.

Note: this question depends on matching the described input/output waveform against four labelled circuit-diagram options (resistor + diode combinations), which could not be reliably reconstructed from the text description alone. Please verify the matching diagram directly against your source PDF's figure before relying on this one.

Q2. Angular momentum of an electron in a hydrogen atom is $3h/\pi$. The energy of the electron is ____ eV.

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- (1) -1.51
- (2) -0.85
- (3) -0.38
- (4) -0.28

✓ **Answer: (3) -0.38**

Solution:

Bohr's angular momentum: $L = nh/(2\pi)$.

Given $L = 3h/\pi = nh/(2\pi)$, so $n = 6$.

Energy $E_n = -13.6/n^2 \text{ eV} = -13.6/36 = -0.38 \text{ eV}$.

Source: NTA B.Tech — 2 April 2026, Shift 1 (official NTA release, physics section) · formulaverse.in